

MS Module 3 Theory

Multi-stream Lehmer random number generators (RNGs)

Multi-stream Lehmer random number generators (RNGs) are designed to produce multiple independent sequences of pseudo-random numbers, essential for parallel computing, reproducibility, and synchronized simulations. These generators extend the classic Lehmer RNG (also known as a multiplicative linear congruential generator) by partitioning its period into distinct streams and substreams.

Multi-Stream Implementations

To support multiple streams:

- **Stream Initialization:** Each stream starts at a seed value spaced sufficiently apart within the generator's period. For example, Steve Park's ANSI C library uses 256 streams, selectable via `SelectStream(s)`.
- **Substream Partitioning:** Streams can be further divided into substreams.
- **State Management:** Libraries like `RngStreams` allow resetting streams to their initial state, advancing to the next substream, or rewinding.

Applications and Considerations

- **Parallel Simulations:** Independent streams enable parallelized Monte Carlo experiments without overlap.
- **Reproducibility:** Seeds and stream indices ensure repeatable results.
- **Statistical Quality:** Properly spaced streams avoid correlation, critical for validity.

Descriptive statistics

- Descriptive statistics are numbers that are used to summarize and describe data.
- They do not involve generalizing beyond the data at hand
- eg. average salaries for various grades of employees in an organisation

Inferential statistics

- Generalizing from our data to another set of cases
- The mathematical procedures whereby we convert information about the sample into intelligent guesses about the population fall under the rubric of inferential statistics
- Inferential statistics are based on the assumption that sampling is random

Data Collection

One of the biggest tasks in solving a real problem. GIGO – garbage-in-garbage-out. Suggestions that may enhance and facilitate data collection:

- **Plan ahead:** begin by a practice or pre-observing session, watch for unusual circumstances
- **Analyze the data as it is being collected:** check adequacy
- **Combine homogeneous data sets,** e.g. successive time periods during the same time period on successive days

- Be aware of data censoring: the quantity is not observed in its entirety, danger of leaving out long process times
- Check for relationship between variables, e.g. build scatter diagram
- Check for autocorrelation
- Collect input data, not performance data

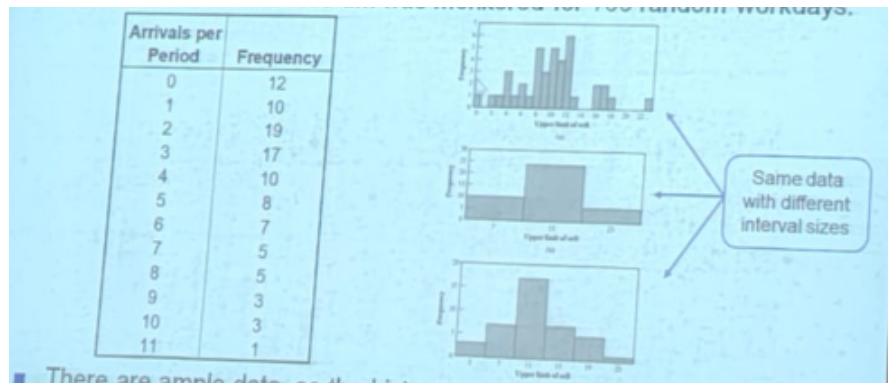
Identifying the Distribution:

- Histograms
- Selecting families of distribution
- Parameter estimation
- Goodness of fit tests
- Fitting a non-stationary process

Data Collection:

- Plan your data collection process
- Always try to find ways that can help you collect data efficiently and accurately
- Collect only data that is useful for your project

Histograms:



There is ample data so the histogram may have a cell for each possible value in the data range.

- Divide the range of the data into intervals
- Label the horizontal axis to conform to the interval selected
- Determine the frequency of occurrences within each interval
- label the vertical axis so that the total occurrences can be plotted for each interval
- Plot the frequencies on the vertical access

A frequency distribution or histogram is useful in determining the shape of a distribution. The number of class intervals depends on:

- The number of observations
- The dispersion of the data
- Suggested: the square root of the sample size

For continuous data:

- Corresponds to the probability density function of a theoretical distribution

For discrete data:

- Corresponds to the probability mass function

If few data points are available combine adjacent cells to eliminate the ragged appearance of the histogram.

Selecting the Family of Distributions:

- Recall if the histogram drawn from your data resembles any kind of statistical distribution.
- Use the physical basis (e.g. usage, discrete or continuous) of the distribution as a guide.
- Use software.
- The exponential, normal, and Poisson distributions are frequently encountered and are not difficult to analyze from a computational standpoint.

A family of distributions is selected based on:

- The context of the input variable
- Shape of the histogram

Frequently encountered distributions:

- Easier to analyze: exponential, normal, and Poisson
- Harder to analyze: beta, gamma, and Weibull

Remember the physical characteristics of the process:

- Is the process naturally discrete or continuous ?
- Is it bounded?

No 'true' distribution for any stochastic input process.

Goal: Obtain a good approximation.

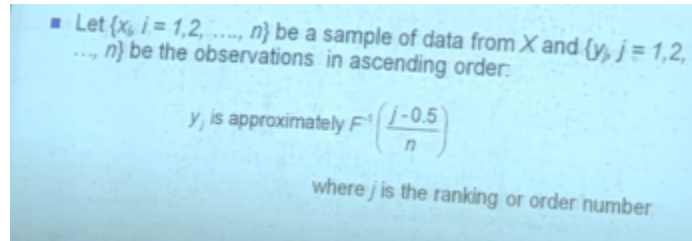
Distribution Types & Physical Basis

- Binomial: successes in n trials.
- Geometric: trials to k successes.
- Poisson: events in time/space.
- Normal: sum of component processes.
- Exponential: time between memoryless events.
- Weibull: time to component failure.
- Uniform: equal probability outcomes.

- Triangular: min, most likely, max known.
- Empirical: resampling from collected data.

Q-Q Plot (Quantile-Quantile Plot)

- Q-Q Plot is a useful tool for evaluating distribution fit
- If x is a random variable with CDF F then the q -quantile of X is the y such that $F(y) = P(X \leq y) = q$, for $0 < q < 1$
 - When F has an inverse, $y = F^{-1}(q)$



- Evaluate the fit of the chosen distributions
- compare the actual values with the values arrived from the chosen distribution
- The nearer to become a straight line the better the accuracy

Consider the following while evaluating the linearity of a q-q plot:

- Observed values never for exactly on the straight line
- the order values are rent and hence not independent unlikely for the points to be scattered about the line
- variance of the extremes is higher than the middle. linearity of the points in the middle of the plot is more important

Q-Q plot can also be used to check homogeneity:

- Check whether a single distribution can represent both sample sets
- plotting the order values of the two data samples against each other

Parameter Estimation:

Population: the collection of data under investigation

Sample: Selected subset

To estimate the parameters of Hypothesized distribution:

- Mean
- Sample Variance

9.3.1 Preliminary Statistics: Sample Mean and Sample Variance

In a number of instances, the sample mean, or the sample mean and sample variance, are used to estimate the parameters of a hypothesized distribution; see Example 9.7. In the following paragraphs, three sets of equations are given for computing the sample mean and sample variance. Equations (9.1) and (9.2) can be used when discrete or continuous raw data are available. Equations (9.3) and (9.4) are used when the data are discrete and have been grouped in a frequency distribution. Equations (9.5) and (9.6) are used when the data are discrete or continuous and have been placed in class intervals. Equations (9.5) and (9.6) are approximations and should be used only when the raw data are unavailable.

If the observations in a sample of size n are $X_1, X_2, \dots, X_n$, the sample mean $\bar{X}$ is defined by

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} \quad (9.1)$$

and the sample variance S^2 is defined by

$$S^2 = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1} \quad (9.2)$$

If the data are discrete and have been grouped in a frequency distribution, Equations (9.1) and (9.2) can be modified to provide for much greater computational efficiency. The sample mean can be computed as

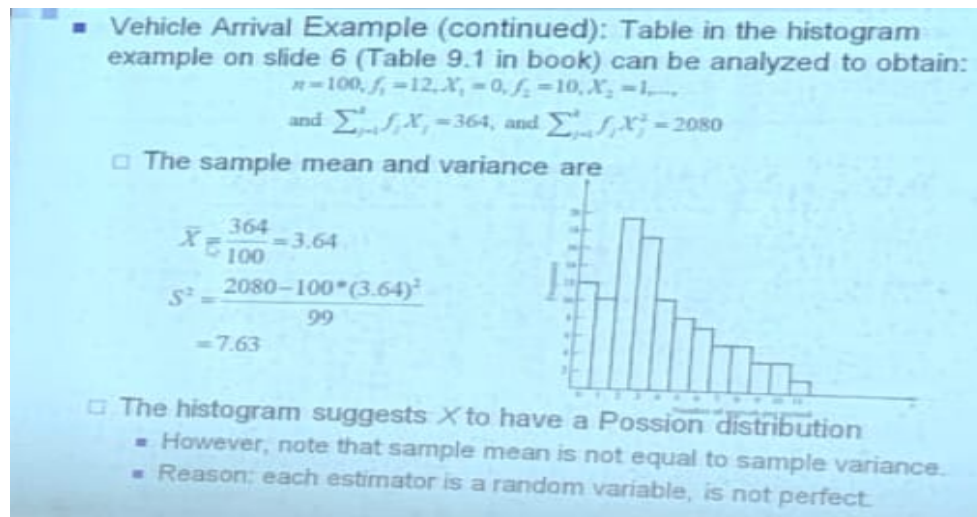
$$\bar{X} = \frac{\sum_{j=1}^k f_j X_j}{n} \quad (9.3)$$

and the sample variance as

$$S^2 = \frac{\sum_{j=1}^k f_j X_j^2 - n\bar{X}^2}{n-1} \quad (9.4)$$

where k is the number of distinct values of X and f_j is the observed frequency of the value X_j of X .

Numerical:



Sample Mean and Sample Variance

- Calculate sample mean ($\bar{X}$) and variance (s^2) from the collected data.

- Based on the distribution chosen, convert the parameters from the sample mean and variance which is (are) used for the distribution

Distribution	Parameter(s)	Suggested Estimator(s)
Poisson	α	$\alpha^{\wedge}=X$
Exponential	λ	$\lambda^{\wedge}=X^{-1}$
Normal	μ, σ^2	$\mu^{\wedge}=X^{-}, \sigma^{\wedge 2}=s^2$

Network of Single-Server Service Nodes

A network of single-server service nodes is a system where multiple service stations (nodes), each with a single server, are interconnected. Customers arrive at the network, move between nodes according to defined routing rules, and receive service at each node before eventually leaving the system.

Key Characteristics

- Single-Server Nodes: Each node serves one customer at a time. If a customer arrives while the server is busy, they must wait in a queue.
- Queueing Discipline: Most commonly, nodes operate under a First Come, First Served (FCFS) discipline, but other disciplines like Last In, First Out (LIFO) are possible.
- Arrival and Service Processes: Arrivals to the network are typically modeled as a Poisson process, and service times at each node often follow an exponential distribution (M/M/1 queues), though generalizations are possible.
- Routing: After service at a node, customers may move to another node, possibly according to probabilistic rules, or exit the system.
- Network Structure: The network can be open (customers eventually leave) or closed (a fixed number of customers circulate among nodes).

Performance and Analysis

- Queue Lengths and Delays: The main performance measures include the average number of customers at each node, average waiting times, and system throughput.
- Heavy Traffic and Correlations: In networks with heavy traffic or correlated service times, the analysis can be reduced to studying an equivalent tandem queue (a series of single-server queues in sequence), sometimes with an added "jitter delay" to account for variations in arrival order.
- Feedback: Some networks allow feedback, where customers may return to previous nodes for additional service, further complicating the analysis.
- Loss Systems: If a node's queue is full, arriving customers may be lost (blocked) from the system, especially in finite-capacity models.